



# Spot the bot: Investigating user's detection cues for social bots and their willingness to verify Twitter profiles

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## ABSTRACT

Detecting social bots is important for users to assess the credibility and trustworthiness of information on social media. In this work, we therefore investigate how users become suspicious of social bots and users' willingness to verify Twitter profiles. Focusing on political social bots, we first explored which cues users apply to detect social bots in a qualitative online study ( $N = 30$ ). Content analysis revealed three cue categories: content and form, behavior, profile characteristics. In a subsequent online experiment ( $N = 221$ ), we examined which cues evoke users' willingness to verify profiles. Extending prior literature on partisan-motivated reasoning, we further investigated the effects of *type of profile* (bot, ambiguous, human) and *opinion-congruency*, i.e., whether a profile shares the same opinion or not, on the willingness to verify a Twitter profile. Our analysis showed that homogeneity in behavior and content and form was most important to users. Confirming our hypothesis, participants were more willing to verify opinion-incongruent profiles than congruent ones. Bot profiles were most likely to be verified. Our main conclusion is that users apply profile verification tools to confirm their perception of a social media profile instead of alleviating their uncertainties about it. Partisan-motivated reasoning drives profile verification for bot and human profiles.

## 1. Introduction

On many social media platforms, users may encounter fully automated social media accounts that emulate human accounts and behavior. These accounts are commonly referred to as *social bots* (Davis et al., 2016). They are created for different purposes, for example, in crisis or news communication (Hofeditz et al., 2019; Hong & Oh, 2020), but have also been linked to malicious activities, including fake rating and reviews, disseminating misinformation, identity fraud, and manipulation of political debates (Bastos & Mercea, 2017; Bessi & Ferrara, 2016; Ferrara, 2017, 2020).

The often hidden automated nature of social bots can make it difficult to detect these accounts by mere eye. However, detecting social bots is crucial for users to judge the trustworthiness and credibility of information on social media. Effective social bot detection is, thus, a defense mechanism for users against possible exertion of influence and manipulation of public opinion. Therefore, current research has focused

on developing profile verification tools (e.g., Botometer<sup>1</sup> or DeBot<sup>2</sup>), which typically indicate the likelihood of a social media account being a social bot (Karataş & Şahin, 2017). They rely on machine learning, using features like meta-data, content, the sentiment of an account, and network structures and activity over time (Lee et al., 2011; Varol et al., 2017).

Nevertheless, we argue that profile verification tools are no panacea to social bot detection. The tools yield different results depending on the detection algorithm (Oberer, Erkollar, & Stein, 2019) and have been criticized for false-positive rates, a narrow consideration of bot activities, and varying accuracy (Grimme et al., 2018; Rauchfleisch & Kaiser, 2020). Detection algorithms are often based on machine learning which is trained through (possibly biased) data sets.

Despite the valid technological concerns about verification tools for research purposes, we argue that it requires more than technical sophistication for profile verification tools to be successful. Their practical relevance also depends strongly on the users' knowledge about social

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<sup>1</sup> <https://botometer.osome.iu.edu/>.

<sup>2</sup> <https://www.cs.unm.edu/~chavoshi/debot/>.

bots and how they actually use these tools (Yang et al., 2019).

With studies investigating the individual user perspective of verification tools being scarce, the main research objective of this work is to examine the user's willingness to make use of these verification tools. Here, we focus on political social bots and suggest that for users to employ a verification tool, they first need to become suspicious of an account. Hence, in a qualitative study, we investigate under which conditions users scrutinize the nature of a social media profile in the sense that they check whether the account is a human or a bot. In other words, we identify and categorize which cues users apply for social bot detection. In a second step, we implement the results of the qualitative study within an experimental study.

With the subsequent experimental study, we aim to understand how users use these cues and employ profile verification tools. Prior results could already identify that one central psychological mechanism affecting social bot perceptions is partisan-motivated reasoning: Twitter profiles displaying political opinions, which are not in line with one's political view, are more frequently perceived as a social bot (Wischniewski et al., 2021; Yan et al., 2020). Building on these results, we suggest that verification tools are applied in a biased manner: Users are more suspicious and therefore are more willing to enter a profile into a verification tool when they disagree with the political opinions shared by a profile and are less willing to do so when they agree with the opinions displayed.

To test this assumption, we are concerned about the *opinion-congruency*, i.e., the (mis)match of the user's and profile's partisanship. Thus, we are not directly interested in the partisanship of a Twitter profile but rather whether users agree with the political opinions expressed by profiles. In the study, we compare different Twitter profiles varying in their *opinion-congruency*. Because it is likely that verification also depends on the extent to which social bots can disguise their automated nature, we additionally compare varying levels of social bot appearance (*type of profile*: bot, ambiguous, human). Here, we are specifically interested in how users perceive the profiles (as opposed to correct or incorrect labeling of a social bots). Therefore, we do not use real social bots in our experimental setting but rely on mockup profiles that vary in their appearance.

We gain first insights into detection cues for social bots and – depending on the type of profile – to which degree users are willing to verify Twitter profiles. The work contributes to the improvement of effective social bot detection by considering psychological biases.

## 2. Background

### 2.1. Effects of partisan-motivated reasoning on information processing online

The question of why specific social media profiles are suspicious of being a social bot is still under-explored. A study by Appling and Briscoe compared cues that users apply for social bot detection with the features used by classification algorithms (including the profile picture, URL and description, username, location, number of followers, followees, and tweets as well as ten randomly selected tweets from a bot and human profile Appling & Briscoe, 2017). Asking participants whether the presented profile was human or non-human-like, the authors found that the essential features for users to detect bots were profile description, username, number of followers, and tweet content. However, it remains open why users found these cues specifically suspicious. The study is also limited by the pre-selection of a set of features. An in-depth exploration of the cues that users use for social bot detection is missing to this date. We close this research gap through a qualitative research approach to identify and characterize the users' detection cues.

Moreover, research on how users actually apply verification tools for social bot detection is scarce. Users can rely on different tools and services, including fact-checking, to scrutinize online information. Hence, we can turn to research on fact-checking to draw possible insights on

how users might use profile verification tools. Fact-checking services assist users in determining truthful online information. However, what is entered into fact-checking services is not necessarily dependent on the perception of truth. Studies have repeatedly shown that individuals' partisanship and political attitudes can bias fact-checking: Users tend to share fact-checks that align with their prior political attitudes (Amazeen et al., 2019; Shin & Thorson, 2017; Walter et al., 2020). However, when individuals are concerned about a specific issue, political partisanship is seemingly less relevant for fact-checking (Rich et al., 2020). Similar biases can be observed regarding the effectiveness of fact-checking. Fact-checking labels that confirm news information do not affect the credibility of a news article but rather increase the perceived quality of the news site (Oeldorf-Hirsch et al., 2020). A meta-analysis revealed that correcting political misinformation is significantly influenced by prior beliefs, ideology, and knowledge (Walter et al., 2020).

These findings for fact-checking can be explained through partisan-motivated reasoning, suggesting that individual reasoning is not always driven to arrive at an accurate but pre-defined conclusion (Kunda, 1990). This *directional cognition* drives reasoning so that prior political attitudes and beliefs (Taber & Lodge, 2006) and group belongings and social identities are confirmed (Cohen, 2003; Kahan et al., 2017; Wischniewski & Krämer, 2021). As a result, individuals more readily accept new information in line with prior attitudes and beliefs but counterargue or reject opposing information (Taber & Lodge, 2006). Against this background, we suspect that, similar to fact-checking, partisan-motivated reasoning may affect the use of social bot verification tools: Prior attitudes and beliefs regarding ones' own partisanship might affect which accounts are entered into profile verification tools in the first place.

Previous research on social bots could already show how partisan-motivated reasoning affects perceptions and behavior on social media: Users who follow social bots, which do not align with their partisanship, become strengthened in their partisanship, which, in turn, can increase political polarization. This effect is especially pronounced for Republicans (Bail et al., 2018). Users are also more willing to engage with these bot accounts (Mosleh et al., 2021). Moreover, accounts with opposing political views are more likely to be perceived as social bots, whereas accounts with similar political views are perceived as less bot-like (Wischniewski et al., 2021; Yan et al., 2020).

In conclusion, these empirical investigations highlight that partisan-motivated reasoning is one key psychological mechanism for using fact-checking services and social bot perceptions. Based on this, we assume that partisan-motivated reasoning also plays a role in profile verification tools of social bots. If incongruent profiles, i.e., profiles which do not express the same partisanship as the user, are more likely to be perceived as social bots, these profiles should also be more likely to be entered into verification tools. In turn, congruent profiles (profiles which share the same partisanship as the user) would be less likely to be entered into verification tools as users are, due to motivated reasoning, less suspicious of these accounts. Hence, verification tools might amplify this bias. We derive the following hypothesis:

- Opinion-incongruent profiles are significantly more likely to be verified than opinion congruent profiles.

In addition to partisan-motivated reasoning, it is likely that different user motivations drive the verification decision. We find that two motivations are most likely to affect the users' decision-making: decreasing the uncertainty about a profile and the confirmation of one's suspicion about a profile. We relate the decreasing uncertainty hypothesis to previous results which indicated that users feel unconfident about detecting social bots without any technical tools (Stocking & Sumida, 2018). Hence, verification tools pose a useful solution to rescind this lack of confidence. However, we think it is also possible that users might apply verification tools in a confirmatory manner, to support their impression of an account. Both motivations, decreasing uncertainty and

confirmation of one's impression, likely result in different behavior. Hence, we explore both hypotheses in the following research question:

- Which type of profile (bot, ambiguous, human) are users more willing to verify?

In particular, if willingness for profile verification would be driven by decreasing uncertainty, then we should find that users are most willing to verify ambiguous profiles as they feel most uncertain about their nature. If the willingness, in turn, is driven by confirmation, users should be more likely to enter an account which they identify as a social bot by themselves.

The local ethics committee of University of Duisburg-Essen approved both studies before data collection. Supplementary material for both studies are publicly accessible on OSF: <https://osf.io/6y3nk/>. All hypotheses were pre-registered on OSF.

### 3. Study 1: Preliminary qualitative study on user knowledge detection cues

#### 3.1. Method

In the first study, we intend to identify which aspects of social media accounts users rely on to become suspicious of a social bot. To investigate this, we conducted a qualitative online study.

##### 3.1.1. Participants

We recruited 30 US citizens through the crowd-working platform Prolific at the beginning of May 2020. Eighteen participants identified as male and twelve as female. Participants' age ranged from 18 to 56 ( $M = 29.20$ ,  $SD = 9.63$ ). Our sample was rather educated (high school diploma  $n = 11$ ; bachelor's degree  $n = 17$ ; master's degree  $n = 2$ ). Most participants spend one to 3 h on average daily on social media, with Facebook, Instagram, and Reddit being the most popular social media platforms.

Out of 30 participants, 17 reported to actively use Twitter. For more insights into the way participants use Twitter we measured their activities through the Twitter Usage Scale (Hughes et al., 2012) on a seven-point Likert scale ranging from 1 (= strongly disagree) to 7 (= strongly agree). Participants answered the following statements about their Twitter usage: "I use Twitter to find and spread information" ( $M = 5.31$ ,  $SD = 0.85$ ); "Twitter is primarily for information" ( $M = 4.69$ ,  $SD = 1.65$ ); "I use Twitter to keep abreast of current events" ( $M = 4.92$ ,  $SD = 1.61$ ); "I use Twitter to keep in touch with friends" ( $M = 4.38$ ,  $SD = 2.36$ ); "I use Twitter because my friends do" ( $M = 4.54$ ,  $SD = 2.18$ ); "Twitter is primarily for socializing" ( $M = 4.46$ ,  $SD = 2.18$ ).

##### 3.1.2. Procedure

We relied on open-ended questions to allow participants to answer in their own words. First, participants were asked what generally makes them suspicious about an account being a social bot. As we could not assume that participants had prior knowledge of social bots, we provided a task presenting Twitter profiles one after another to them. In this task, they were asked to which degree they believed that the presented profile was a social bot or a human (rating from (1) social bot to (100) human). Participants did not receive feedback on this task as we did not want to prime them for specific cues. After the task, participants were asked about the strategies they used to evaluate the profiles and what made them think that the profile is run by a social bot or a human (e.g., specific cues or behavior). At the end of the survey, participants were debriefed and received 4.70\$ as compensation for participation. On average, participants needed 14:57 min to fill out the study.

##### 3.1.3. Qualitative content analysis

Using the software MAXQDA 2018, we applied a top-down, concept-driven approach and performed a structuring content analysis (Mayring,

2014). Structuring content analysis is characterized by a top-down approach in which the categories are determined before data collection as opposed to other content analysis approaches (e.g., conventional content analysis) that intend to describe phenomenon and avoid pre-determined categories (Hsieh & Shannon, 2005). We pre-defined one main categories on *detection cues of social bots* for our coding scheme. In the first coding cycle, we applied descriptive coding and in-vivo coding to capture participants' voices (Saldaña, 2013). Then, codes were summarized and abstracted into higher-level codes, resulting in our final coding scheme (see supplementary material). Besides the pre-defined category, a new category emerged from our data: *detection cues of human accounts*, which captured the characteristics of a human account. After the first coding cycle, the whole body of data was recoded, and new subcodes were created. Marginal codes applied less than four times in the whole data body and only used by one participant were removed from the initial coding scheme.

We calculated Cohen's  $\kappa$  (Brennan & Prediger, 1981), considering a value of 90% rater's agreement as a base, i.e., the textual segments coded by each rater had to match at least at a rate of 90% to be considered as an agreement. A second rater coded seven randomly selected participants' survey answers at the lowest subcode level. This procedure was applied two times as the first round revealed disagreements in the coding scheme. We discussed the coding scheme and the disagreements in-depth. Based on these discussions, we updated the categories and specified the meaning and distinction between specific categories (e.g., canned replies (pre-determined replies) vs. same content). The second round of coding resulted in an inter-rater reliability score of  $\kappa = 0.79$ , which can be considered sufficient (Landis & Koch, 1977).

#### 3.2. Results

In the following, we present our results regarding the categories. Participants will be referred to as "P" and their respective number. Our analysis revealed four categories of social bot detection cues (Table 1): *content and form*, *behavior*, *profile characteristics*, and *gut feeling*. The most prominent cue category for social bot detection was *content & form*. This category included commercial content, "promoting things like books or products" (P18) and random or nonsensical content, "saying random things, "which was sometimes accompanied by negative sentiment, e.g., "[...] I saw a bot posting stupid questions that were unrelated." (P23). Furthermore, social bots did not post original or personal content, such as P4 reported, "[...] if there was no reference to their personal life in any way, it seemed like a script." Instead, two participants stated that the *content and form* of social bots were persuasive, P3: "The bot accounts are focusing on pushing a narrative.". The *content and form* of social bots were characterized by their (flawed) language. For instance, P10 stated: "Terrible diction/grammar, formatting of words and phrases [...]", or P11: "[...] the semantics seemed odd [...].". The social bots were also characterized by a lack of intelligence, posting "stuff that didn't have thought" (P18). Finally, a minority of participants reported that links to other websites were an important cue for participants, "a lot of random links" (P1).

The *behavior* of a Twitter profile was seen as another essential cue category. Within this category, homogeneous behavior was the most frequent cue. Homogeneous behavior included posting the same content focusing on one topic and canned replies, and regular and constant behavior. Moreover, we identified cues that we coined *repeating behavior* (e.g., "only retweets" (P10)) and "no behavior" (e.g., "no retweets" (P1)). From the examples, it was evident that for participants, the cue was more about the homogeneity of behavior than the type of content. Other cues in this category included engagement with others, as P11 highlighted: "I felt like social bots typically had really low engagement in their posts [...].", or as P23 explained more specifically: "[...] the bot does not comment anything or retweet someone with a message." Finally, high volumes of posts and tweets, as well as high activity, indicated a bot, P14: "[

**Table 1**

First and second level codes for of cues of social bot detection.

| 1st code level          | Description                                                                        | No. of participants | 2nd subcode level                                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Content and form        | Refers to the content that is (typically) posted by the social Bots                | 24                  | <ul style="list-style-type: none"> <li>- Randomness/nonsensical content</li> <li>- Advertisement/commercial content</li> <li>- No original/personal content</li> <li>- Persuasion</li> <li>- Content form</li> </ul> |
| Behavior                | Refers to the behavior that is realized by the social bots/typical for social bots | 24                  | <ul style="list-style-type: none"> <li>- Engagement with others (reactions, comments)</li> <li>- High volume of posts/tweets/activity (spamming)</li> <li>- Homogeneity</li> </ul>                                   |
| Profile characteristics | Refers to different aspects of the (bot) account                                   | 12                  | <ul style="list-style-type: none"> <li>- Profile picture</li> <li>- Age of the account</li> <li>- Username</li> <li>- Location</li> <li>- Self-description</li> <li>- Friends/follower/followee/ratio</li> </ul>     |
| Gut feeling             | Refers to vague and subjective perceptions                                         | 3                   |                                                                                                                                                                                                                      |

...] they retweet a lot of posts.” and “just retweets of the same things often many times in a day or short time span.” (P16).

The cue category *profile characteristics* referred to profile picture, a profile description, age of the account (e.g., P22 “a very new account” indicates a social bot), username, location, number of the follower as well as follower-followee ratio, e.g., P11 “[...] it would be even more likely to be run by a bot if the account had a really high amount of followers and almost no engagement.”, or P22”. Concerning the cue category *gut feeling*, a minority of three participants reported that “something just felt off” (P26) and that they “can’t really pinpoint specific features” (P11), indicating that they relied on an overall impression of the Twitter profile to determine whether the account was a social bot. This overall impression was unspecified and “weird” (P29).

While we did not ask explicitly about the cues for human detection, many participants mentioned the cues they used to determine a human account to contrast them to the social bot detection. We were able to identify the nearly identical categories (content and form, behavior, and profile characteristics) for the detection of human accounts and found that human accounts were often characterized with opposite content and form (e.g., sense of personality and emotion) and behavior (e.g., heterogeneity). Thus, social bot accounts missed a human touch. These unexpected categories of cues for human account detection highlighted that users detect social bots through contrasting human and bot accounts.

### 3.3. Discussion

Our qualitative study showed that social bot detection is performed by contrasting the account at hand with human accounts. Human accounts were characterized by expression of personality, i.e., expression of emotions, humor, opinions, and thoughts. In contrast to human *content and form*, social bot *content and form* were characterized by poor language (e.g., wrong grammar and spelling), randomness, i.e., nonsensical content or not contextual content, and a lack of personality and intelligence. Regarding behavior, repetitive behavior and no-behavior were considered crucial social bot characteristics, i.e., accounts that only performed one action (e.g., retweets only) or never performed specific actions (e.g., no comments or engagement in terms of liking content). This suggests that, regarding *content and form* and behavior, inflexibility and homogeneity are ascribed to social bots. All in all, the qualitative analysis allows us to explore and characterize three concrete categories of detection cues for social bots: (1) *content and form*, (2) *behavior*, (3) *profile characteristics*. Moreover, some participants reported relying on a vague gut feeling.

This study presents an exploration of possible detection cues, i.e., which cues might be generally important to users. However, the study does not address which cues would evoke participants to use a verification tool. Hence, we investigated this question further in a second

study. Thus, we intended to explore the following research question:

- Which detection cues contribute to the willingness for profile verification?

## 4. Study 2: How do users verify Twitter accounts?

### 4.1. Method

In the qualitative study, we identified the three detection cue categories. We derived a list of detection cues for each category based on participants’ answers (Table 2) and quantified the detection cues in the second experimental study. Moreover, the experiment builds on former results on the role of motivated reasoning in the perception of social media accounts (Wischniewski et al., 2021; Yan et al., 2020) and investigates how it can affect the willingness for profile verification. We pre-registered the study on OSF before data collection: <https://osf.io/93a64>.

#### 4.1.1. Participants

We recruited 221 US American Twitter users through the crowdsourcing platform Prolific. Participants’ age ranged from 18 to 75 (M

**Table 2**

List of detection cues based on the qualitative study.

| Category                | Cue                                                                                                                                                                                                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Content and form        | Degree of personality<br>Degree of emotionality<br>Degree to which the content is persuasive<br>Degree to which the content contains balanced views<br>Degree of creativity<br>Degree to which the content of this profile is repetitive.<br>Number of links to other websites<br>Grammar and spelling<br>Choice of words |
| Behavior                | Use of posts<br>Use of retweets<br>Use of comments<br>Degree to which the profile focuses on one topic                                                                                                                                                                                                                    |
| Profile characteristics | Profile picture<br>Profile description<br>Location<br>Age of the account (creation time of the account)<br>Username<br>Number of followers/followees (and its ratio)                                                                                                                                                      |

Note. We asked participants the following questions: “Did you consider [profile characteristics, the profile’s behavior, the content of this profile] for this decision? “[Yes, No] If yes: “How important were the following cues for your decision? Hint: We are interested in which cues you considered for your decision. Which cues made you suspicious of the profile, and which cues assured you of its nature? This could also include the absence of certain cues.”



= 34.25,  $SD = 12.44$ ). One hundred seventeen participants identified as female, 96 as male, and five as non-binary (three not shared). Regarding ethnicity, 153 identified as White, 27 Black or African American, 22 Hispanic or Latino, and 13 Asian or Pacific Islander, six identified as other or mixed ethnicities. The sample was rather educated. Most participants ( $n = 96$ ) held a bachelor's degree, while 34 achieved a master's degree and four a doctorate. Seventy-three participants had a high school diploma, and three people had less than a high school diploma.

In our sample, 178 participants identified as Democrats and 43 as Republicans. Most of them were very familiar with Twitter and used the platform several times a day ( $n = 143$ ; once a day,  $n = 33$ ; at least once a week,  $n = 33$ ; less than once a month,  $n = 12$ ). For more insights into why our participants used Twitter we, in line with our first study, measured their activities on the platform through the Twitter Usage Scale (Hughes et al., 2012). Participants answered the following statements about their Twitter usage: "I use Twitter to find and spread information" ( $M = 4.04$ ,  $SD = 2.07$ ); "Twitter is primarily for information" ( $M = 4.31$ ,  $SD = 2.10$ ); "I use Twitter to keep abreast of current events" ( $M = 5.03$ ,  $SD = 1.88$ ); "I use Twitter to keep in touch with friends" ( $M = 3.45$ ,  $SD = 2.14$ ); "I use Twitter because my friends do" ( $M = 3.10$ ,  $SD = 2.14$ ); "Twitter is primarily for socializing" ( $M = 3.58$ ,  $SD = 2.05$ ).

#### 4.1.2. Study design and procedure

We conducted an online experiment using a  $3 \times 2$  within-subject design with "type of profile" (three levels: bot, ambiguous, human) and "opinion-congruency" (two levels: congruent, incongruent) as independent variables.

Participants were first asked for their knowledge about social bots and their experience with profile verification tools. Independently of their prior knowledge and experience, we first gave a definition of social bots (to ensure all participants have the same understanding<sup>3</sup>). Second, we informed the participants about profile verification tools, defining them as "tools that examine the Twitter activity of a profile to determine whether this profile is run by a social bot or a human user by giving a score. If the score is high, it is more likely that the profile is a social bot". Then, we introduced a scenario: Participants were asked to imagine meeting the following Twitter profiles in the study in their everyday Twitter use. Eighteen Twitter profiles were presented to the participants, one after another in a random order, and they were requested to inspect them in detail. After each Twitter profile, we asked the participants for their willingness to verify the presented profile with a verification tool, the detection cues they used to determine the bot, whether the participants viewed the profile as bot-like or human-like, and how certain participants were about their appraisal. If participants did not choose any cues from our list, they needed to answer an open question about their profile verification decision ("What did you consider to make your decision?"). After presenting the Twitter profiles, we asked for demographics and control measures (see 4.1.4 Measures). In the end, participants were debriefed and received 3.73\$ for participation. The average completion time was about 36 min.

#### 4.1.3. Mockup twitter profiles

We created 18 mockup Twitter profiles (three for each condition; Fig. 1) that varied in their type and political opinion expressed. Twitter profiles were designed to appear as either clearly run by a social bot/human or not clearly bot-like/human-like (ambiguous) in their nature. To create the profiles, we followed the two-step procedure by

<sup>3</sup> We gave the following definition using bullet points for readability: "Social bots are automated online accounts that communicate more or less autonomously; typically operate on social media sites, such as Twitter; are programmed to forward, like or comment on specific topics like the weather, sport results, but also political issues; can be programmed by companies to disseminate advertisement within social networking sites; can openly state that they are automated accounts."

Wischniewski et al. (2021). As a first step, we created a pre-selection of self-identified social bot profiles on Twitter to serve as examples for social bot accounts and verified human profiles to serve as examples for human profiles. Because even self-identified social bots might still represent human profiles, we selected only these profiles from the pre-selection, which displayed previously found bot-features such as the timing of posting behavior, shared content, and displayed profile picture (see Wischniewski et al., 2021 for more details). The final selection of profiles served as a template for our mockup profiles which we manipulated, in a last step, to either express more Democrat or Republican content. We chose to include actual and timely content shared either by Democrats or Republicans on Twitter for political opinion. This procedure also ensured that the content was timely, increasing the external validity of the study. Through the procedure, we ensured that the mockup profiles can represent real social bots.

#### 4.1.4. Measures

We assessed the willingness of profile verification through a 5-point Likert scale ("What is the likelihood of you entered this profile into a verification tool?"; 1 = very unlikely, 3 = undecided, 5 = very likely) and detection cues which were informed by the prior qualitative results (Table 2). We included the measurement for the botness of the Twitter profiles. Participants needed to judge the degree to which they think the presented Twitter profile represents a social bot or a human account (slider from (1) bot to (100) human). A low score indicated a bot-like perception, while a high score indicated a human-like perception. We further asked for the certainty of this judgment ("How certain do you feel towards this judgment?", slider: 1 (very uncertain) – 100 (very certain)).

Several control variables were included. We measured the social bot experience through a single item ("How much do you think you know about 'social bots'?") on a 5-points-Likert scale with the following labels: 1 = A great deal/I'm an expert on this topic; 2 = A lot/I have read quite a lot about them; 3 = Some/I have some knowledge about them.; 4 = A little/I have only heard of them.; 5 = None/I have no idea what they are (Yan et al., 2020). Furthermore, we assessed whether participants are familiar with profile verification tools and have used them before, how much time they spend on social media and on Twitter, and whether they identify as democrat or republican.

#### 4.2. Results

We followed our pre-registered strategy of analysis with one exception: We conducted three ANCOVAs using type of profile and opinion-congruency as within-subject factors to analyze the effects on profile verification and negative and positive sentiment and include the control variables. The analysis considered mean-centered values of age, education, time spent on Twitter, time spent on social media, familiarity with profile verification tools, and social bot knowledge as covariates. The results of our pre-registered analysis can be found in the supplementary material on OSF. We applied Bonferroni correction to post-hoc comparisons.

A one-way repeated measures ANOVA for botness using type of profile as a factor revealed a main effect, ( $F(1.94, 426.40) = 308.04$ ,  $p < .001$ ,  $\eta_p^2 = 0.58$ , using Greenhouse-Geisser correction). Pairwise post-hoc comparisons showed that botness perception was the highest for human profiles ( $M = 75.14$ ), over ambiguous profiles ( $M = 56.06$ ), and bot profiles ( $M = 38.39$ ). The differences between all type of profiles were significant indicating that our manipulation of the Twitter profiles was successful.

##### 4.2.1. Willingness for profile verification

To address H1 which suggested that incongruent profiles would be more likely to be verified than congruent profiles, a paired-samples t-test was conducted to compare the willingness for profile verification of congruent and incongruent profiles (across all types of profiles). We found a significant difference for congruent ( $M = 2.88$ ,  $SD = 0.64$ ) and incongruent ( $M = 3.20$ ,  $SD = 0.67$ ) profiles;  $t(220) = -6.97$ ,  $p < .001$ ,

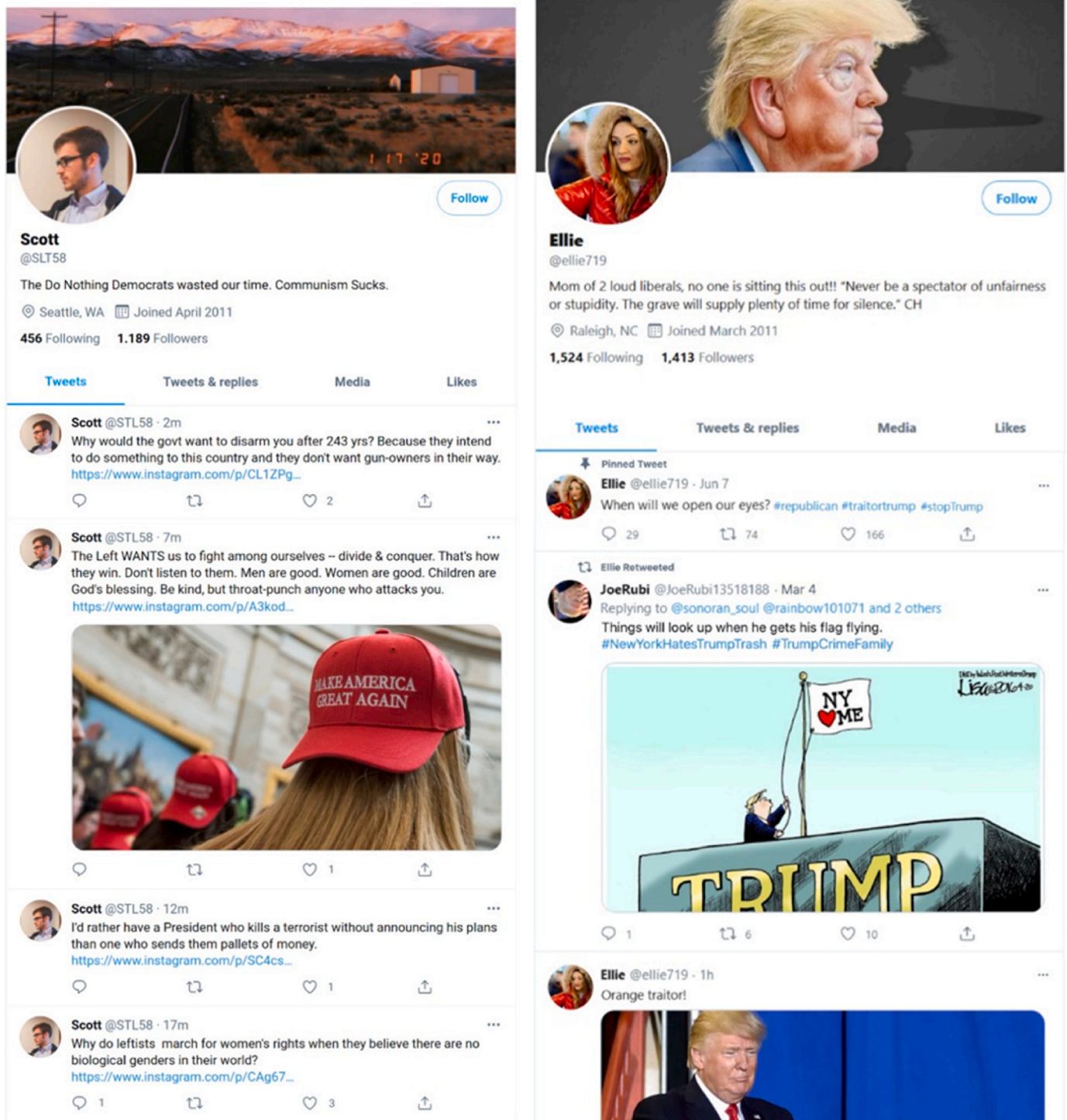


Fig. 1. Examples for a bot republican (left) and a human democrat (right) mockup Twitter profile.

95% CI for mean difference  $[-0.41, -0.23]$ , Cohen's  $d = -0.47$ ). The effect can be considered moderate, confirming our hypothesis that incongruent profiles are more likely to be verified by users than congruent ones.

Moreover, the two-way ANCOVA revealed a large significant main effect for *type of profile*,  $F(1.94, 416.83) = 269.56$  (using Greenhouse-Geisser correction),  $p < .001$ ,  $\eta_p^2 = 0.56$ , and *opinion-congruency*,  $F(1, 220) = 51.98$ ,  $p < .001$ ,  $\eta_p^2 = 0.20$ , on willingness for profile verification. Additionally, the analysis showed a significant interaction effect for *type of profile*  $\times$  *opinion-congruency*,  $F(1, 220) = 21.95$ ,  $p < .001$ ,  $\eta_p^2 = 0.093$

Table 3

Descriptive of willingness for profile verification and certainty for all conditions.

| Type of profile | Congruency  | Willingness for profile verification $M$ ( $SD$ ) | Certainty $M$ ( $SD$ ) |
|-----------------|-------------|---------------------------------------------------|------------------------|
| Human           | Congruent   | 2.06 (.88)                                        | 76.53 (18.39)          |
|                 | Incongruent | 2.59 (1.01)                                       | 72.47 (19.24)          |
| Ambiguous       | Congruent   | 3.01 (.95)                                        | 70.20 (18.56)          |
|                 | Incongruent | 3.11 (.93)                                        | 70.17 (18.53)          |
| Bot             | Congruent   | 3.57 (.86)                                        | 70.32 (17.88)          |
|                 | Incongruent | 3.90 (.87)                                        | 73.53 (19.35)          |

(Table 3). Post-hoc tests revealed that for bot and human profiles, incongruent profiles were more likely to be verified than congruent ones. This effect, however, did not apply to ambiguous profiles. Here, opinion-congruency did not affect the *willingness for profile verification*. The covariates *age* ( $F(1, 220) = 9.27, p = .003, \eta_p^2 = 0.04$ ) and *education* ( $F(1, 220) = 8.93, p = .003, \eta_p^2 = 0.04$ ) were significantly related to congruency: older participants and more educated participants were more willing to verify profiles.

This result highlights two aspects: Firstly, partisan-motivated reasoning can be found for bot and human profiles but was absent for ambiguous ones. Secondly, willingness for profile verification was seemingly more driven by confirmation rather than uncertainty as participants were more willing to verify bot profiles than ambiguous and human ones.

To further scrutinize whether willingness for profile verification was driven by confirmation or by the decrease of uncertainty, we conducted an additional analysis regarding *certainty*. First, we calculated correlations between *willingness for profile verification*, *botness*, and *certainty* (Table 4).

The results showed a significant negative correlation between *willingness for profile verification* and *certainty* for the human condition and no significant correlations for the ambiguous and bot conditions. A similar pattern can be found for the correlations for *botness* and *certainty*. Here, results showed a significant positive correlation for the human condition but no significant correlations for the ambiguous and bot conditions. In addition, we found significant negative correlations between *willingness for profile verification* and *botness* for all conditions.

Taken together, the results suggest that *willingness for profile verification* was apparently not driven by uncertainty. If this were the case, then a low *certainty* would be related to a high *willingness for profile verification*, i.e., a significant negative correlation for all conditions. However, the results show that *certainty* was unrelated to *willingness for profile verification* for the ambiguous and bot conditions.

The correlations between willingness for profile verification and botness, in turn, indicate that confirmation might be the motivational driver as a negative correlation could be found in all conditions. Thus, in all conditions, a low botness (i.e., user perceived the respective profile as a social bot) was associated with a high degree of willingness for profile verification.

As the correlation results did not account for congruency, we further conducted a two-way repeated measure ANCOVA for *certainty* using *type of profile* and *opinion-congruency* as within-subject factors. The analysis showed a significant main effect for *type of profile*,  $F(1.94, 414.78) = 11.65$  (using Greenhouse-Geisser correction),  $p < .001, \eta_p^2 = 0.05$ , and a significant interaction effect for *type of profile*  $\times$  *opinion-congruency*,  $F(2, 440) = 9.26, p < .001, \eta_p^2 = 0.04$ .

Pairwise post-hoc comparisons showed that the difference between bot ( $M = 71.96$ ) and human profiles ( $M = 74.50$ ) was significant, as well as the difference between human and ambiguous profiles ( $M = 70.18$ ). Bot and ambiguous profiles did not differ significantly, i.e., participants were equally certain about ambiguous profiles (Fig. 2b). Furthermore, for human profiles, participants felt significantly more certain about congruent profiles ( $M = 76.53$ ) than incongruent ones ( $M = 72.48$ ).

Interestingly, we found the reverse pattern for bot profiles: Participants were significantly more certain about incongruent bot profiles ( $M = 73.53$ ) than congruent ones ( $M = 70.32$ ). Yet, participants were more willing to verify incongruent profiles (Fig. 2). This supports that willingness for profile verification is driven by confirmation rather than uncertainty.

#### 4.2.2. Detection cues

To address our research question of which cues contribute to willingness for profile verification, we performed a linear mixed effect analysis using R version 4.1.1 and lme4 package (Bates et al., 2015). We included *willingness for profile verification* as dependent variable and all detection cues as fixed effects (without interaction term). Moreover,

participants and the 18 profiles were included as random effects in the model.<sup>4</sup> Willingness for profile verification was significantly predicted by the degree to which the content contains balanced views, the number of links to other websites, the degree to which the content of a profile is repetitive, and the degree to which the content focuses on one topic. Furthermore, the age of an account, the degree of personality and emotionality as well as the posting and commenting behavior contributed negatively to the willingness for profile verification. (ICC = 0.27, marginal  $R^2 = 0.102$ , conditional  $R^2 = 0.343$ ; Table 5).

## 5. General discussion

Focusing on political social bots, we investigated which cues users apply to detect social bots and to which degree they are willing to verify Twitter profiles. Concerning the detection cues, in study 1, we identified the detection cues categories: content and form, behavior, and profile characteristics. As a qualitative study, study 1 did not provide insights into which cues evoke participants to use profile verification tools. Thus, study 2 was designed to investigate which of the identified detection cues contribute to the willingness for profile verification. Here, we asked which detection cues participants consider when making their assessment about a profile's nature. We found that participants' assessment primarily concerns salient holistic cues including, for example, the degree of emotionality or personality. These holistic cues rely on an overall impression of Twitter profiles rather than the specific cues that can be assessed quickly by a mere glance (e.g., profile characteristics). Against the results of study 1, we can assume that a high degree of personality and emotionality indicate a human. Thus, these detection cues contribute negatively to the willingness for profile verification. Homogeneity (through the focus on one topic or repetitive content) indicates social bots can contribute negatively to the willingness for profile verification. In this light, the age of the account is most likely crucial as users' overall impression of Twitter profiles also relies on a temporal dimension. In other words, without a certain age of a profile, it would not be possible to express personality or emotionality.

### 5.1. Motivated verification for bot and human profiles

Concerning the willingness for profile verification, we found that users were more willing to verify bot-like profiles and employ profile verification in a *biased manner*: The user's partisanship impacted the willingness of profile verification in a way that profiles which did not share the same political attitude were more likely to be verified. Thus, we found that partisan-motivated reasoning impacts the willingness for profile verification tools which is in line with previous findings on social bot perceptions (Wischniewski et al., 2021; Yan et al., 2020). Moreover, our participants felt less sure about the nature of bot-like profiles that shared the same political opinion than profiles that did not. It seems that the fact that an account shared a user's view made it more difficult for a user to identify the account as a social bot which supports the motivated reasoning hypothesis. In turn, this holds important implications for social media users.

The results imply that social bot profiles that share the same partisanship are less likely to be scrutinized regarding their nature. Consequently, users might examine their content less critically. Thus, these kinds of profiles can affect users without hindrance. Users could become more vulnerable to manipulative social bot accounts and the content they might spread. This gives misinformation or conspiracy theories spread by automated accounts a loophole.

Moreover, we observed that users were more willing to verify bot-

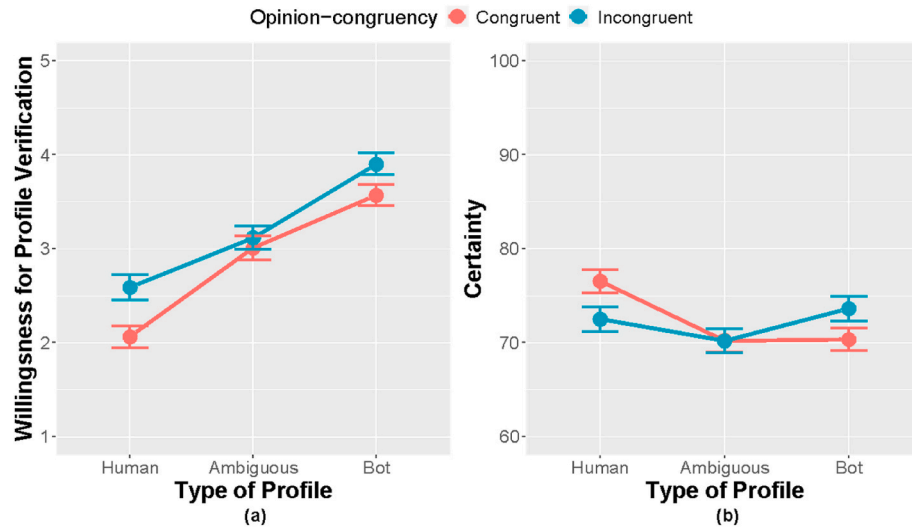
<sup>4</sup> The answers to open-ended questions were neglected for this analysis (59 answers of which 36 could be assigned to one of the three main categories, however, we did not have any data on how relevant these cues were for the detection).



**Table 4**

Correlation matrix for willingness for profile verification, botness, and certainty for each type of profile.

|       | PVHum   | PVAmb   | PVBot   | BTHum   | BTAmb  | BTBot | CTHum   | CTAmb   | CTBot |
|-------|---------|---------|---------|---------|--------|-------|---------|---------|-------|
| PVHum | 1       |         |         |         |        |       |         |         |       |
| PVAmb | 0.44*** | 1       |         |         |        |       |         |         |       |
| PVBot | 0.19**  | 0.28*** | 1       |         |        |       |         |         |       |
| BTHum | −.68*** | −.22*** | −0.06   | 1       |        |       |         |         |       |
| BTAmb | −0.17** | −.70*** | −0.20** | .30***  | 1      |       |         |         |       |
| BTBot | 0.08    | −0.02   | −.65*** | 0.11    | .25*** | 1     |         |         |       |
| CTHum | −.34*** | −0.16*  | −0.09   | 0.41*** | 0.1    | −0.02 | 1       |         |       |
| CTAmb | −0.08   | −0.11   | −0.03   | 0.17*   | 0.03   | −0.08 | 0.73*** | 1       |       |
| CTBot | −0.06   | 0.06    | −0.02   | 0.12    | −0.12  | −0.13 | 0.65*** | 0.64*** | 1     |

\*Note: PV = willingness for profile verification, BT = Botness, CT = Certainty, Hum = Human, Amb = Ambiguous, significance levels: \* $p < .05$ , \*\* $p < .01$ , \*\*\* $p < .001$ .**Fig. 2.** (a) Main and interaction effects for (a) willingness for profile verification and (b) certainty.**Table 5**

Results of linear mixed model for willingness for profile verification.

| Cue category            | Fixed effects        | Estimates | CI         |
|-------------------------|----------------------|-----------|------------|
| Profile characteristics | Intercept            | 2.86 ***  | 2.54–3.19  |
|                         | Profile picture      | −0.01     | −0.06–0.03 |
|                         | Profile description  | −0.01     | −0.06–0.03 |
|                         | Location             | −0.03     | −0.07–0.02 |
|                         | Age of the profile   | −0.08 *** | −0.12–0.04 |
|                         | Username             | 0.02      | −0.03–0.06 |
|                         | Number of Followers  | 0.01      | −0.03–0.06 |
| Content                 | Personality          | −0.15 *** | −0.20–0.09 |
|                         | Emotionality         | −0.11 *** | −0.16–0.06 |
|                         | Persuasiveness       | −0.04     | −0.09–0.00 |
|                         | Balanced view        | 0.08 ***  | 0.03–0.12  |
|                         | Creativity           | 0.04      | −0.01–0.09 |
|                         | Number of Links      | 0.10 ***  | 0.06–0.14  |
|                         | Grammar and Spelling | −0.01     | −0.06–0.03 |
| Behavior                | Choice of Words      | −0.05     | −0.10–0.00 |
|                         | Repetitive content   | 0.20 ***  | 0.16–0.25  |
|                         | Posting behavior     | −0.06 **  | −0.11–0.02 |
|                         | Retweeting behavior  | 0.05 *    | 0.01–0.09  |
|                         | Commenting behavior  | −0.08 *** | −0.12–0.04 |
|                         | Focus on one topic   | 0.18 ***  | 0.14–0.22  |

Note. \* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

like profiles over ambiguous and human ones despite being more similarly uncertain about ambiguous profiles. This contradicts the assumption that users employ verification tools to reduce uncertainty about a profile. Instead, the results suggest that users use profile verification tools to *confirm* their perceptions rather than decrease their uncertainty about a profile's nature. This is problematic as users only identify those accounts which they already suspect to be social bots, whereas accounts

that appear ambiguous or human might not be verified by users in the first place. Consequently, these kinds of social bots remain largely undetected by users.

Another explanation for this result concerns the users' trust in the performance of profile verification tools. Users might not believe that the tools perform better than they did, and therefore, do not deem it worthwhile to use the tools on accounts that are not clearly bots. However, we note that we did not ask for users' trust in the performance of profile verification tools. All in all, our results suggest that profile verification tools may fail their primary objective in supporting individual users in detecting social bots as the tools can only support users when they are already suspicious of an account but not beyond that.

Interestingly, we found partisan-motivated reasoning only for bot and human profiles but *not* for ambiguous ones. In other words, bot-like and human-like profiles that were not in accordance with the users' partisanship were more likely to be verified by participants. However, for ambiguous profiles, whether the profile was in accordance with the partisanship did not affect the willingness for profile verification. This is somewhat surprising given that we argue that ambiguous profiles do require more deliberate thinking to make a judgment about verification which, in turn, could lead to a higher reliance on the individual's partisanship. Consequently, we should find motivated reasoning for ambiguous profiles as well.

To understand these results, we rely on previous findings of the effects of deliberation on motivated reasoning, which suggest that deliberative reasoning reduces the effects of motivated reasoning (Bago et al., 2020; Pennycook & Rand, 2019). We assume that participants were not incentivized to deliberately think about their decision for bot-like and human-like profiles because the profiles clearly represented bot and human profiles. Hence, motivated reasoning could affect their



perceptions. In turn, ambiguous profiles did neither show clear signs of automation nor clear signs of a human user. Consequently, participants had to engage in deliberative reasoning to decide about the nature of an account.

### 5.2. Implications for social bot detection

Our results have several implications for the issue of social bot detection. Against the background of our findings, we conclude that profile verification tools that allow individual users to detect social bots might not be effective as the tools are seemingly rather used to confirm prior bot perception. In our view, profile verification tools (e.g., Botometer or DeBot) need to consider these user biases, for example, by informing users about them. Furthermore, we think it is necessary to further investigate how users interact with such profile verification tools. How do users, for instance, interpret the scores that are provided to them? To which extent are these interpretations biased by the confirmatory way in which users apply profile verification? And, most importantly, to what extent can these scores correct users' perceptions if they falsely assumed a profile as a social bot (especially when the profile does not share the same opinion)? Due to the bias in profile verification, we suggest that *platform-based social bot detection* is more appropriate than profile verification tools for individual users. In other words, we argue that social media platforms should handle and be responsible for social bot detection removing malicious social bots automatically from the platform.

In addition, insights concerning the detection cues found in our work, which users employ, could be implemented in already existing literacy programs. Similarly, detection cues could be included in places where platforms inform users about platform manipulation and spam policies (e.g., <https://help.twitter.com/en/rules-and-policies/platform-manipulation>) to assist users.

### 5.3. Limitation and future work

Our study mainly relied on self-reports in an experimental setting in which participants were asked to which extent they were willing to enter a profile into a verification tool. The study did not involve any verification through an actual tool. Users might use profile verification to alleviate uncertainties about the nature of a social media profile in a real-life setting. Furthermore, we note that our results are specific to partisanship in the US context and the platform of Twitter. The number of characters limits content on Twitter. Thus, detection cues might be different for other platforms, which allow for more content. Therefore, we suggest that future work could investigate the real-life application of profile verification tools in a more realistic setting and for different social media platforms. Furthermore, we suggest investigating the topic of transparent and benign social bots: How does transparency in the form of self-identifying information affect users' perceptions of social bots? Addressing these questions could support the trustworthiness of benign social bots in critical settings such as crisis communication. Theoretically, our study leaves open why we found the observed pattern of opinion-congruency. More precisely, did opinion-congruency reduce users' ability to detect social bots by, for example, decreasing attention (as the content does not trigger a threat), by increasing heuristic processing (reliance on partisan cues), or maybe both?

## 6. Conclusion

With a focus on Twitter, this work studied which cues users apply to detect social bots and to which degree they are willing to verify profiles. Confirming prior studies, the results show that users know little about social bots and have negative views towards them. Furthermore, content, behavior, and profile characteristics are essential categories for cues. Social bot profiles are perceived as characterized by inflexibility and homogeneity: Repetitive content and the focus on one topic make a

Twitter profile bot-like. Moreover, profile verification tools are used in a *confirmatory manner*. This means that users are more willing to verify profiles that are already perceived as bot-like. Furthermore, partisan-motivated reasoning impacts the willingness for profile verification. Users are more willing to verify profiles that do not share the same political views. Consequently, social media users might be more vulnerable to the possible manipulation of social bots that share the same political view. We conclude that profile verification tools for individual users might not support users effectively in detecting social bots and suggest a shift to platform-based social bot detection.

### Credit author statement

Thao Ngo: Conceptualization, Methodology, Formal analysis, Investigation, Writing - Original Draft, Visualization. Magdalena Wischniewski: Conceptualization, Methodology, Formal analysis, Writing - Original Draft. Rebecca Bernemann: Conceptualization, Methodology, Writing - Review & Editing, Visualization. Martin Jansen: Writing - Review & Editing. Nicole Krämer: Conceptualization, Writing - Review & Editing, Supervision, Funding acquisition.

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### Declaration of competing interest

None.

### Data availability

Data is available on OSF.

### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chb.2023.107819>.

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